Probability and Statistics B - Practical Sheet 3

Getting started:

1. Open **RStudio** (do not use the base R application).
2. From the menu, select **File** → **New File** → **R Script**. This will create a blank script where you can type and run your R commands.
3. Immediately save your script with a meaningful name, e.g., `Practical1.R`, using **File** → **Save As**. Choose a location you can easily find later.
4. Type your commands in the script and run them using:
   - **Ctrl + Enter** (Windows) or
   - **Cmd + Enter** (Mac)
5. When you have your answers, copy the relevant output (not the entire script) into a Word document.

Important: Always keep your R script saved. This is good practice and ensures you have a record of your work.

1. A farmer has been offered a new fertiliser. Before placing a large order, he uses it on a random sample of 5 'test' plots. The results are:

   ```
   15.9, 14.5, 12.7, 13.8, 14.2
   ```

   (a) Suppose he knows that the measurements for the fertiliser follow a $\mathrm{N}(\mu, 1)$ distribution, where μ is unknown. Use the method of moments to calculate a 95% confidence interval for the population mean μ. (R will provide the mean but do the other calculations manually).

   (b) Suppose now it is known that $\mu = 15$, but the farmer doesn't know this. He has a lot of time on his hands so decides to perform the experiment many times. Simulate this by generating many samples using:

   ```
   x1 = rnorm(200, 15, 1)
   x2 = rnorm(200, 15, 1)
   x3 = rnorm(200, 15, 1)
   x4 = rnorm(200, 15, 1)
   x5 = rnorm(200, 15, 1)
   m1 = cbind(x1, x2, x3, x4, x5)
   ```

   i. View the first two samples by typing:
   ```
   m1[1,]
   m1[2,]
   ```
   ii. Calculate the 200 sample means using:
   ```
   sample.mean = apply(m1, 1, mean)
   ```
   iii. Calculate the lower end of the 95% CI using:
   ```
   ci.lower = sample.mean - 1.96/sqrt(5)
   ```
   iv. Calculate the upper end using:
   ```
   ci.upper = sample.mean + 1.96/sqrt(5)
   ```
   v. View the 95% CIs by typing:
   ```
   data.frame(ci.lower, ci.upper)
   ```

Do any of these confidence intervals **not** contain the population mean $\mu = 15$? Count the number that do not contain μ by using:

```
dd = data.frame(ci.lower, ci.upper)
dd[with(dd, order(ci.lower)), ]
```

Is this what you would expect for 95% confidence intervals?

2. (a) The data below are the number of hours between charging on a mobile phone:

```
45.8 41.1 55.9 46.6 57.0 45.0 58.5 46.7 49.3
52.7 54.9 48.5 40.4 44.4 51.0 44.2 59.1 46.9
50.7 43.7 41.7 52.8 60.5 38.5 60.4 53.8 47.3
50.2 58.8 50.7
```

   (b) The data below are the number of hours spent watching television per week:

```
23.1 15.9 21.0 26.0 25.1 14.7 24.2 16.6 18.2
16.5 20.7 15.3 17.7 19.1 22.7 21.9 14.6 26.3
25.8 9.4 17.0 21.2 17.9 24.7 21.1 17.2 19.1
22.7 24.0 24.7 22.5 8.3 2.5 30.4
```

   (c) The data below are the number of miles travelled to and from work each day:

```
3.7 14.3 11.0 26.5 5.2 4.8 24.2 16.9 8.2
26.5 40.7 5.3
```

   For each dataset, find 95% and 99% confidence intervals for the population mean. For example use:

```
mean(x1) + c(1,-1) * qnorm(0.025) * sd(x1)/sqrt(30)
```

3. A set of data consists of the following 15 values. Suppose that the data corresponds to a random sample from a population

```
12.4 16.7 20.5 17.0 10.6 14.9 16.8 21.6 17.7 10.2 15.1 18.2 19.3 21.0 15.8
```

   (a) Scan the data into an R vector called x. Obtain a 95% confidence interval using:

```
t.test(x)
```

   (b) For a 99% confidence interval for the underlying population mean. Use the same command as before but add the subcommand `conf.level=0.99`.

   (c) Obtain a boxplot and stem-and-leaf diagram for `x` and justify whether the t distribution is appropriate.

   (d) Compare the 95% CI that you would obtain using the normal approximation. You can do this manually or in R the following code can be used.

```
mean(x) - 1.96 * sd(x)/sqrt(15)
mean(x) + 1.96 * sd(x)/sqrt(15)
```

4. (From Tutorial Sheet 5) A laboratory experiment is performed to compare two methods of evaluating the coefficient of thermal expansion of a polymer. Sample of eighteen varying grades of a polymer are used, nine with method A and nine with method B, yeilding the following results:

Method A	Method B
6.8	7.2
8.4	8.3
7.9	8.3
6.5	7.5
9.5	9.5
7.0	7.7
9.1	9.4
7.5	7.8
9.8	10.3

Set up two vectors in R, one called "methodA" and the other "methodB". Use R to determine a 95% confidence interval for the difference between the population means for method A and Method B:

```
t.test(methodB, methodA, var.equal = TRUE)
```

5. An investigator was studying people's ability to recognise faces when they are shown in black-and-white (b-w) and in colour (col). She had 21 volunteers and split them into two groups: ine group of 10 were shown the faces in black-and-white and the other group of 11 were shown them in colour. The variable measured was the average time (secs) taken to recognise the faces. The resulting data are given in the following table:

```
b-w: 1.7 1.5 1.6 1.9 1.2 1.9 1.6 1.2 1.2 1.0
col: 1.3 1.8 1.6 1.2 1.7 1.5 1.0 1.4 1.5 1.0 1.1
```

   (a) Use R to find a 95% confidence interval for the ratio of two population variances for the samples above. If they are entered as vectors "bw" and "col" the command is:

   ```
   var.test(bw, col)
   ```

   (b) Obtain a 95% confidence interval for the difference in population means. How does the confidence interval in (a) help you justify this answer?

   (c) Obtain boxplots of the samples above and comment further on how this helps justify your answer for the difference in the population means.

6. Here are some sample results. Use "scan" to copy them to an R vector.

```
241 251 258 246 202 246 246 301 260 228 275 266 264 250 258 265 240 227 271 306
269 252 242 274 222 244 218 228 247 202 287 279 242 262 268 226 303 214 242 233
```

   Obtain a 95% confidence interval to determine if it is plausible that these data come from a population with mean 260.